

Bradford Protein Assay

Overview

A colorimetric method for quantifying protein concentration in a sample.

A. Materials

Bradford reagent (e.g., Coomassie Brilliant Blue G-250-based)	As required
BSA stock solution (2 mg/mL)	As required
ddH ₂ O	As required for dilution
Protein sample(s) to be quantified	As required
96-well microplate (or cuvettes, if using a spectrophotometer instead of plate reader)	1
Microplate reader (or spectrophotometer set to 595 nm)	1
Multichannel or single-channel pipettes	As required

B. Procedure

1. Prepare the BSA standard curve by diluting 2 mg/mL BSA stock with ddH₂O into six tubes/wells, per the table below (100 μ L total each):

	600 μ g/mL	500 μ g/mL	400 μ g/mL	300 μ g/mL	200 μ g/mL	100 μ g/mL
BSA	30	25	20	15	10	5
ddH ₂ O	70	75	80	85	90	95

2. Dilute your protein sample(s) with ddH₂O if needed, so the expected concentration falls within the 100–600 μ g/mL range of your standard curve.
3. Add 100 μ L of each standard into separate wells of the microplate. Include one well with 100 μ L ddH₂O only as a blank.
4. Add 196 μ L of Bradford reagent + 4 μ L of samples to every well, only add bradford reagent to blank. Mix gently (e.g., by pipetting up and down or gentle

shaking) — avoid making bubbles, since they interfere with the absorbance reading.

5. Incubate at room temperature for 5–10 minutes, protected from light. Color development (brown to blue) happens during this window.
6. Measure absorbance at 595 nm using a microplate reader, zeroing/blanking against the ddH₂O-only well.
7. Plot the standard curve: absorbance (y-axis) vs. known BSA concentration (x-axis) for your six standards, and fit a linear trendline.
8. Use the trendline equation to calculate your unknown sample concentrations from their measured absorbance values.
9. Run standards and samples in duplicate or triplicate if possible, and average the results for better accuracy.

General Formula

Total volume needed = (number of wells) × (volume per well) × 1.1

The ×1.1 (10% extra) accounts for pipetting loss, dead volume in tubes, and reagent that sticks to pipette tips

1. Bradford reagent

Volume needed = (total number of wells: standards + blank + samples) × 100 µL × 1.1

Example: 6 standards + 1 blank + 3 samples = 10 wells

→ 10 × 100 µL × 1.1 = 1,100 µL (1.1 mL) of Bradford reagent

2. BSA stock (2 mg/mL) for the standard curve

This is fixed regardless of how many samples you run, since it only depends on your 6-point standard curve: Volume needed = (30 + 25 + 20 + 15 + 10 + 5) µL × 1.1 = 105 × 1.1 = ~116 µL, round up to ~120 µL of 2 mg/mL BSA stock. If running standards in duplicate or triplicate, multiply by 2 or 3 first, then apply the ×1.1.

3. ddH₂O

Two separate uses, add them together:

For standard dilutions: (70 + 75 + 80 + 85 + 90 + 95) µL × 1.1 = 495 × 1.1 = ~545 µL

For the blank well(s): 100 µL × (number of blank wells) × 1.1

For diluting samples (if needed): depends on your sample's starting concentration — use $C_1V_1 = C_2V_2$ to figure out how much water dilutes your sample into the 100–600 µg/mL range, then add that volume too.

4. Protein sample

Volume needed = 100 µL × (number of replicate wells for that sample) × 1.1

If you're testing 3 samples in triplicate: $100\ \mu\text{L} \times 3\ \text{replicates} \times 3\ \text{samples} \times 1.1 = \sim 990\ \mu\text{L}$ total sample volume needed (split across your 3 actual samples, so $\sim 330\ \mu\text{L}$ each before replicating).

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